Project Title
(Recipe Finder)

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by
2520030123 — POLISETTY SREEHARSHA
2520030331 — ARNAV CHITIKENA
2520030573 — HARSHITH MADDINENI
Team Number: 13
Section: 2

Under the guidance of

Dr. S. Vinay Kumar, Associate Professor,

Department of Computer Science and Engineering

(DEEMED TO BE UNIVERSITY)
BACHUPALLY

Hyderabad-500090, Telangana, India.

ABSTRACT

Problem statement

Finding suitable recipes can be difficult when users search based on available ingredients rather than the exact recipe name. Traditional recipe searching methods may require checking a large number of recipes one by one, making the process inefficient. Therefore, there is a need for an efficient recipe finder that can quickly search, match, and recommend recipes based on user requirements.

The proposed Recipe Finder is a Java-based application that allows users to search recipes by recipe name, available ingredients, cuisine, and category. The system also provides spelling correction for incorrect search queries, displays complete recipe and ingredient information, and supports recipe selection based on factors such as cooking time, budget, rating, and nutritional values. The application uses structured recipe data stored in CSV files, including, to organize and retrieve recipe information efficiently.

The project is implemented using Java in Visual Studio Code with a console-based user interface and CSV file-based data storage. Advanced Data Structures and Algorithms concepts are applied to improve searching and recommendation efficiency, including Knuth–Morris–Pratt (KMP), Rabin–Karp, Edit Distance using Dynamic Programming, Bipartite Matching, Knapsack Optimization, and Randomized Hashing. These techniques enable efficient pattern matching, ingredient matching, spelling correction, meal optimization, and fast recipe retrieval, demonstrating the practical application of DSA concepts in a real-world recipe search system.

Signature of the Guide